The sharp A_p bound for the Hardy–Littlewood maximal operator Simon Gustafsson

Notation.

- Cubes are closed with sides parallel to the axes. A cube Q has side length $\ell(Q)$, and $3Q$ is the cube with the same centre and side length $3\ell(Q)$.
- $Q(x,r) := \{y : |y-x|_\infty \le r/2\}$ is the cube centred at x with side length r, where $|z|_\infty := \max_k |z_k|$.
- $v(E) := \int_E v\,dy$ for measurable $v \ge 0$ and measurable $E \subseteq \mathbb{R}^d$. In particular $|E|$ is the Lebesgue measure of E.

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \ge 0$ that is not almost everywhere zero.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \le p < \infty$. The weighted L^p space is*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(y)|^p\, w(y)\, dy < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(y)|^p\, w(y)\, dy\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \ge 0 : w(\{|f| > t\}) = 0\} < \infty$.

Definition 3 (Maximal operators)**.**

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f|\, w\, dy, \qquad M^c_w f(x) := \sup_{r>0} \frac{1}{w(Q(x,r))} \int_{Q(x,r)} |f|\, w\, dy.$$

For $w \equiv 1$ these are M and M^c.

Definition 4 (A_p characteristic)**.**

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w\, dy\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{p-1},$$

and $w \in A_p$ if $[w]_{A_p} < \infty$.

Definition 5 (Dual weight)**.** *$\sigma := w^{-\frac{1}{p-1}}$, so that* $[w]_{A_p} = \sup_Q \dfrac{w(Q)}{|Q|}\left(\dfrac{\sigma(Q)}{|Q|}\right)^{p-1}$.